

K-ART-4L

A Reproducible Expert-Annotated Benchmark for Layered Computational Analysis of Korean Fine Art

통합 논문 초안 | 한국 미술의 계층적 계산 분석을 위한 재현 가능한 전문가 주석 벤치마크

Anonymous Submission

Target: AI Datasets and Benchmarks / Computer Vision for Art Analysis

Submission-integrity note. This manuscript separates verified claims from planned experiments. It does not report numerical performance, annotator agreement, or generation quality before the corresponding benchmark runs are executed. The final submission must replace the reporting-plan tables with frozen, reproducible outputs and archive the exact data, code, seeds, and environment used to produce them.

Abstract

Modern vision-language models inherit art concepts from corpora and taxonomies that underrepresent Korean fine art. The resulting problem is not only a shortage of images; it is a mismatch between the formal categories used by Korean art practice and the labels used by computational systems. We introduce K-ART-4L, a benchmark specification and data-governance pipeline that represents Korean fine art through four explicitly separated but non-independent layers: iconography, technique, materiality, and color. The benchmark is designed around artist-, artwork-, collection-, and source-disjoint evaluation; expert-verified multilabel annotations; bilingual terminology retrieval; and auditable licensing metadata. We further define a controlled comparison of pooled baselines and Layer-Query Attention (LQA), a lightweight adapter that assigns layer-specific queries to frozen visual tokens. Unlike prior drafts, we do not treat attention weights as explanations, translation gaps as cultural gaps without controls, or an automatic recognizer as a sufficient judge of generated cultural validity. We specify paired bilingual retrieval tests, held-out human-coded layer fidelity, counterbalanced expert evaluation, bootstrap confidence intervals, mixed-effects models, multiplicity control, and preregistered failure criteria. The paper also establishes a strict scope boundary: K-VALUE price estimation, K-ART LENS product evaluation, and synthetic Tobit experiments are downstream applications rather than evidence for this benchmark. The result is a verifiable protocol for measuring whether multimodal systems can recognize Korean art using categories defined and audited by domain experts.

Keywords: Korean art; cultural heritage AI; dataset and benchmark; vision-language models; expert annotation; retrieval; materiality; reproducibility; data governance

1. Introduction

Computational art analysis has benefited from large collections such as WikiArt, Rijksmuseum, OmniArt, and open-access museum data. These resources support style, genre, metadata, and retrieval tasks, but their dominant ontologies are not neutral. They tend to foreground categories such as oil on canvas, chiaroscuro, impasto, and Christian iconography. Korean fine art often requires additional or differently structured descriptions: auspicious motifs, embodied brush actions, absorption and bleeding on hanji, pigment behavior, and color systems that combine perceptual appearance with symbolic direction and meaning.

When these categories are absent, a model may map a Korean concept to a visually nearby but art-historically different English concept. Such a failure should not be inferred from an isolated qualitative example or from a lower score in Korean. It requires a benchmark with explicit labels, controlled translations, leakage-resistant splits, expert agreement, and uncertainty estimates. K-ART-4L is designed to provide that benchmark.

1.1 Research questions

- RQ1. Can existing vision-language and visual encoders recognize Korean fine-art properties when evaluated on expert-defined layer labels rather than generic style labels?

- RQ2. Does performance differ between Korean aesthetic terminology and professionally matched English queries after translation quality and query difficulty are controlled?
- RQ3. Does Layer-Query Attention improve layer-specific recognition over capacity-matched pooled and multi-head baselines?
- RQ4. Are the resulting representations layer-specific in a measurable predictive sense, and are their attention maps faithful to the evidence used by the classifier?

1.2 Contributions

Contribution	What is released or tested	What is not claimed
Dataset protocol	A four-layer taxonomy, multilabel schema, provenance fields, license routing, expert annotation procedure, and leakage audit.	Not that four layers are universally independent or that the dataset represents all Korean art.
Benchmark	Recognition, paired bilingual retrieval, and optional generation tracks with frozen splits and reporting rules.	Not that a single scalar proves cultural understanding.
Method comparison	Capacity-matched pooled heads, independent heads, adapters, and LQA under identical seeds and splits.	Not that attention weights are explanations without faithfulness tests.
Governance	Image-level rights metadata, recognition-only versus generation-eligible routing, and model/data documentation.	Not that C2PA establishes authorship, ownership, or authenticity.

1.3 Scope boundary

K-ART-4L is a recognition and retrieval benchmark. It does not estimate market price, authenticate artworks, replace an appraiser, or certify compliance with any appraisal standard. K-VALUE is a separate downstream research program whose identification problem requires transaction and selection data not supplied by this benchmark. K-ART LENS is an application prototype, and DLthon2 is an implementation study; neither is used as evidence of K-ART-4L validity.

2. Related Work and Problem Definition

2.1 Art datasets and cultural coverage

Existing art datasets established the value of large-scale visual recognition and metadata prediction. However, dataset scale does not solve an ontology mismatch. K-ART-4L therefore treats label design, annotation provenance, and split integrity as first-class benchmark objects. The aim is not to replace general art datasets, but to make a culturally specific evaluation slice that can expose failures hidden by broad genre or style accuracy.

2.2 Four-layer representation

For an artwork image x , K-ART-4L stores a structured label $y=(y_{\text{icon}}, y_{\text{tech}}, y_{\text{mat}}, y_{\text{color}})$, optional bilingual captions, uncertainty or missingness indicators, and source-level provenance. The four layers are operational axes, not an ontological claim that the properties are statistically independent. Their correlations are measured and reported.

Layer	Operational definition	Annotation unit	Known risk
Iconography	Motif, category, narrative or symbolic meaning when supported by evidence.	Artwork and region when a local motif is identifiable.	Interpretive disagreement; long-tail motifs.
Technique	Observable or documented making action, brush method, ink behavior, or construction process.	Artwork, region, or documented process.	Technique may be unidentifiable from a single reproduction.
Materiality	Support, pigment family, surface behavior, absorption, bleeding, relief, or texture.	Artwork plus high-resolution crop or RTI subset.	Image quality and lighting confounds.
Color	Perceptual palette plus culturally relevant symbolic relations when documented.	Artwork-level palette and graph labels.	Color is partly deterministic and must not dominate aggregate scores.

2.3 Formal tasks

T1: multi-layer recognition. Given x , predict the multilabel tuple y with hierarchical and uncertainty-aware evaluation. T2: paired bilingual retrieval. Given a Korean query q_{ko} or a professionally matched English query q_{en} , retrieve images from the same fixed pool. T3: optional culturally conditioned generation. This track is not required for the core benchmark and must not be reported unless a held-out human-coded evaluation is completed.

3. Dataset Construction

3.1 Release tiers and unit of analysis

Each record receives a stable artwork_id, image_id, source_id, collection_id, artist_id when known, creation-period metadata, image dimensions, preprocessing hash, rights status, label provenance, and a release tier. Image files are redistributed only when the license permits redistribution. Otherwise, the benchmark distributes URLs or institutional access instructions together with metadata and annotations, subject to the source terms.

Tier	Use	Release rule
A: redistributable	Training, validation, and evaluation when the license permits.	Image, metadata, annotations, and license record.
B: reference release	Recognition and retrieval through institution-approved access.	URL or access pointer, metadata, annotations, hashes; no unauthorized image redistribution.
C: restricted	Metadata-only or recognition-only evaluation.	No generation use; access and deletion procedures documented.

The dataset card must list the exact license or permission for every image. A public API or a public web page is not itself evidence that derivative training, redistribution, or generation is permitted. The Korean Copyright Commission describes fair use through purpose, work type, amount, and market impact factors; the benchmark therefore records a rights decision rather than asserting a blanket legal exception. [18]

3.2 Sampling and label support

The benchmark core includes only labels with sufficient support for the frozen split. A proposed inclusion rule is at least 50 training instances and 20 test instances per benchmark class after artist- and artwork-disjoint partitioning. Long-tail motifs remain available as metadata or open-vocabulary evaluation, but they are not converted into unstable macro-F1 classes. Materiality crops and RTI images are separated from artwork-level recognition so that a texture reference cannot leak into the artwork test set.

The final paper must report actual counts, not target counts. It must include a class-support table, source and collection composition, artist concentration, period composition, image resolution distribution, and the number of records removed by duplicate and rights audits.

3.3 Leakage-resistant splits

- Normalize filenames, metadata, and image dimensions; compute pHash and perceptual near-duplicate clusters.
- Assign all scans, crops, catalog variants, and near-duplicates of an artwork to one split.
- Assign all works by an artist to one split for the primary artist-disjoint evaluation.
- Report collection-disjoint and source-disjoint sensitivity splits when feasible.
- Reserve a temporal holdout only for tasks where creation or publication time is meaningful; do not mix temporal and random splits without labeling them separately.

3.4 Expert annotation and adjudication

Graduate students may propose candidate labels, but benchmark labels are produced by at least two independent domain experts for core items. A 20-30% overlap is labeled by a third expert to estimate agreement and identify ambiguous categories. Annotators work from a versioned codebook that defines positive, negative, unknown, and not-visible states. Adjudication is performed after the independent labels are frozen and follows a written rule; it is not allowed to silently replace disagreement with a single convenient answer.

Component	Required evidence
Annotator qualification	Role, domain experience, training, conflict-of-interest statement, and compensation.
Codebook	Definitions, positive/negative examples, hierarchy, distance matrix, ambiguity rules, and revision history.
Agreement	Krippendorff's alpha or multilabel agreement with confidence intervals; hierarchical agreement where applicable.
Gold labels	Raw independent labels, adjudication log, missingness, and label-distribution release.

4. Models

4.1 Frozen visual backbone

Let a frozen encoder map image x to patch-token features H in $\mathbb{R}^{(n \times d)}$. We compare CLIP, SigLIP, and DINOv2 using their published checkpoints and exact preprocessing. We do not fine-tune the backbone in the primary comparison unless a separate budget-matched experiment is reported.

4.2 Layer-Query Attention

For each layer l in {icon, tech, mat, color}, LQA uses a learnable query q_l and layer-specific projections. The layer representation is:

$$z_l = \text{softmax}((q_l W_l^Q)(H W_l^K)^T / \sqrt{d_k}) H W_l^V \quad (1)$$

A classifier g_l predicts the labels for layer l . The final paper must specify d , d_k , d_v , number of heads, parameter count, trainable parameter fraction, FLOPs, memory, and latency. The claim that LQA adds less than one percent of backbone parameters is not accepted without this calculation. A shared-projection LQA variant should be included because layer-specific projections can obscure whether the gain comes from queries or capacity.

4.3 Retrieval alignment

For T2, the text side is defined explicitly. A layer-conditioned query encoder maps the Korean or English query to the same embedding space used for image retrieval. Positive pairs are image-query pairs supported by the annotation record; hard negatives share the artwork period or broad genre but differ in the target layer. If the implemented system uses only prompt-based zero-shot retrieval, the paper must call it zero-shot diagnosis and must not claim that LQA learned bilingual alignment.

4.4 What counts as layer specificity

We use task-specificity rather than the stronger and potentially misleading term disentanglement. A representation z_l is layer-specific when it predicts its own layer better than matched-capacity representations and when cross-layer probes do not show an equivalent representation of every other layer. This is an empirical property, not a consequence of naming four query vectors.

5. Evaluation Protocol

5.1 T1 recognition metrics

We report per-layer macro-F1, micro-F1, mAP, hierarchical F1, balanced accuracy where appropriate, Brier score, and expected calibration error. Rare classes are reported with support counts and precision-recall curves. Aggregate scores are macro-averages over layers, not image-weighted averages that allow deterministic color features to hide failures in iconography or technique.

5.2 T2 paired bilingual retrieval

Each query pair is written or verified by three qualified bilingual annotators. Reverse translation and semantic-equivalence checks are recorded. The Korean and English versions use the same image pool and the same target label. We report R@1, R@5, R@10, MRR, nDCG@10, per-layer scores, and a paired gap with bootstrap confidence intervals. The gap is described as a language-conditioned retrieval gap unless controlled experiments support a cultural interpretation.

$$\text{Gap}@k = R@k(q_{\text{en}}) - R@k(q_{\text{ko}}) \quad (2)$$

A normalized sensitivity form may also be reported, but it is never used alone because a small English denominator can make ratios unstable. Query difficulty, token count, translation confidence, and class support are included in a mixed-effects sensitivity model.

5.3 T3 generation track

Generation is an optional extension. If included, each condition tuple must contain held-out combinations not seen during training. Generation uses fixed seeds and reports distributional metrics only as secondary evidence. Layer fidelity is measured by an independent recognizer and by human coding on a held-out sample. An automatic recognizer trained on the same labels is not sufficient evidence of cultural validity.

5.4 Human evaluation

Expert ratings use a counterbalanced, blinded, repeated-measures design. The primary endpoints are layer-specific plausibility, cultural dissonance, and pairwise preference. Authentic-versus-generated discrimination is a descriptive diagnostic, not a success objective. Rater and item are random effects; the paper reports the ICC specification, effect estimates, confidence intervals, and missing-rater rules.

6. Statistical Analysis Plan

6.1 Model comparisons

Every primary model is trained with at least five predeclared random seeds. Test predictions are frozen before analysis. Differences are evaluated with paired bootstrap resampling over test artworks, not independent-sample tests that ignore repeated images or shared queries. We report mean, standard deviation, 95% confidence interval, effect size, and the proportion of seeds in which the method wins.

6.2 Multiple comparisons and power

The primary endpoints are fixed before training: T1 macro-F1 averaged over the four layers and T2 paired retrieval gap at R@10. Secondary layer-wise, backbone-wise, and ablation comparisons are corrected with Holm or Benjamini-Hochberg procedures. A simulation-based power analysis uses the planned class supports, number of artworks, expected seed variance, and paired effect size; the paper does not justify power solely by the number of images.

6.3 Ablation matrix

Ablation	Question	Required control
Pooled head vs. four heads	Does layer-specific output structure matter?	Same backbone, parameter budget, optimizer, and seeds.
Four heads vs. LQA	Does token-level query attention add value?	Report trainable parameters and FLOPs.
Shared vs. layer-specific projections	Is the gain due to query design or capacity?	Equalize parameter count.
Uniform vs. uncertainty weighting	Does loss weighting improve rare layers?	Predefine weighting and monitor calibration.
Single-task vs. multitask	Does joint learning help or cause negative transfer?	Same class support and early-stopping rule.
Attention faithfulness	Do attended regions support the prediction?	Deletion/insertion and shuffled-query controls.

6.4 Failure criteria

The benchmark is successful even when a model fails, provided the failure is measured reproducibly. We predefine that no cultural-gap claim is accepted if the bilingual gap disappears under translation-quality controls, if artist leakage is detected, if confidence intervals include the practical null effect, or if the human-coded evaluation contradicts the automatic judge. Negative results are retained in the release and discussion.

7. Reproducibility and Governance

7.1 Release package

- Versioned dataset manifest with artwork_id, image hash, source, license, split, label provenance, and deletion contact.
- Complete codebook, raw annotations, adjudication logs, label distance matrix, and class-support report.
- Data loaders, preprocessing code, duplicate audit, split generator, training scripts, evaluation scripts, and environment lockfile.
- Checkpoint hashes, model cards, dataset card, random seeds, hardware, wall-clock time, and failure logs.
- One-command evaluation that reproduces the tables and figures from frozen predictions.

7.2 Provenance and authenticity language

Content Credentials or C2PA manifests may document the provenance of digitization and generated outputs. They do not establish that an artwork is authentic, that a person owns it, or that a label is art-historically correct. The benchmark therefore uses provenance metadata as an audit signal and keeps authenticity claims outside the task definition. [19]

7.3 Cultural and ethical governance

Religious imagery and living artists require explicit handling. Recognition-only records are quarantined from generation experiments. Cultural restrictions, takedown procedures, contributor consent, and institutional review are documented at image level. The benchmark does not infer that a public-domain image is unrestricted for every downstream use. Generated outputs must carry a clear AI-source disclosure, and the benchmark prohibits presenting generated material as an authentic historical artwork.

8. Limitations

First, the four layers are a practical schema and not a complete theory of Korean art. Iconography and technique can be interpretive, while materiality is often unobservable from a low-resolution reproduction. Second, artist-disjoint evaluation measures transfer but may underestimate performance in collection-management settings where the same artist appears repeatedly. We therefore report both transfer and in-domain sensitivity splits without conflating them. Third, bilingual retrieval is affected by language technology and translation quality even after controls. Fourth, the optional generation track cannot eliminate cultural appropriation risk through metrics alone. Finally, no benchmark score establishes artistic value, market price, authenticity, or professional appraisal validity.

9. Conclusion

K-ART-4L proposes a verifiable benchmark for a specific failure mode in multimodal art AI: the inability to recognize culturally meaningful formal categories that are absent or distorted in dominant art corpora. The central contribution is not a claim that four labels solve Korean art, nor that a single attention map explains a model. It is a disciplined evaluation object: expert-defined labels, leakage-resistant splits, bilingual query controls, capacity-matched baselines, independent human checks, uncertainty reporting, and auditable governance. Once the benchmark run is completed, the resulting evidence will support either a measured cultural representation gap, a narrower language-conditioned retrieval finding, or a transparent negative result. Any of these outcomes is scientifically preferable to claiming cultural understanding from placeholders, uncontrolled translations, or self-referential judges.

Appendix A. Pre-submission completion gate

Gate	Pass condition	Evidence required
Data reality	Actual released counts replace every target count.	Manifest, class-support table, source audit.
Annotation quality	Agreement and ambiguity are measured before model testing.	Raw labels, alpha/CI, adjudication log.
Leakage	No artwork, scan, crop, or near-duplicate crosses splits.	Audit report and hash clusters.
T1	All primary baselines and LQA run with predeclared seeds.	Frozen predictions and evaluation logs.
T2	Bilingual pairs are professionally verified and difficulty-controlled.	Translation audit and paired bootstrap.
T3	Only if independent human-coded evaluation is complete.	Held-out labels, rater protocol, mixed-effects output.
Governance	Every record has a rights route and takedown path.	License matrix, dataset card, policy.
Narrative	No numeric placeholder, unresolved count, placeholder quote, or unsupported first claim remains.	Final manuscript search and claim ledger.

Appendix B. Downstream work excluded from this benchmark

K-VALUE requires a separate, identified market estimand, lot-level sold and unsold records, known or interval-censored reserve information, a defensible selection equation, temporal evaluation, group-conditional uncertainty, and professional review. Synthetic censoring in DLthon2 can test implementation but cannot validate real market selection correction. K-ART LENS can be evaluated as an application after the benchmark labels and retrieval judgments exist. These projects should cite K-ART-4L as a data or benchmark dependency, not as evidence that their own downstream claims have been established.

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